MIND OVER MATTER

Spacing

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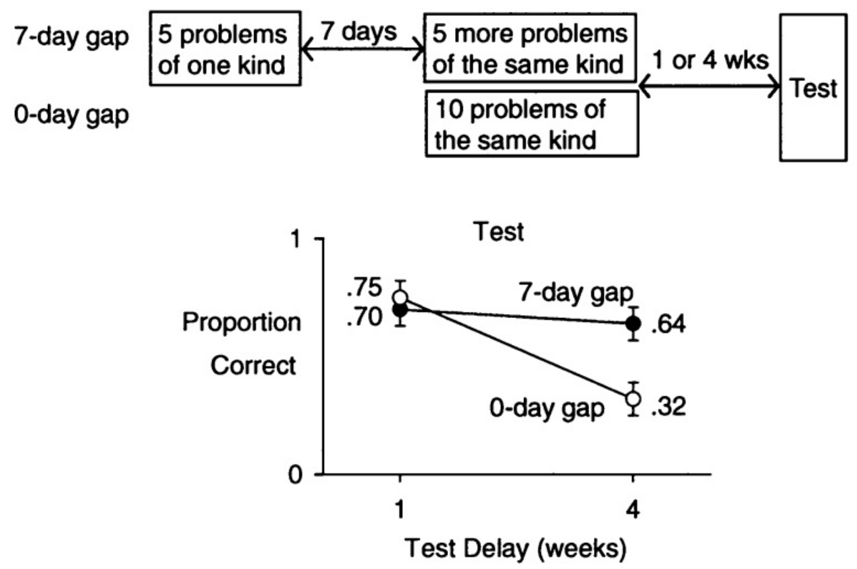
Spacing involves distributing learning over time rather than massing it into a single block. This is an intervention that improves memory by forcing the brain to engage in active memory reconstruction. Spacing applies to learning, studying, AND assessment. Retrieval practice is not only our favorite teaching and learning strategy, but also among the best ways to take full advantage of spacing. Each time a student is forced to "re-load" information after a delay, the neural pathways for that information are strengthened.

Spacing can be particularly tough for our students because...

- Spacing requires organization, both for teachers (deciding the order of lessons and units) and for students (putting together study plans, etc.)

- Learning with spaced practice can feel frustrating, slow, and ineffective, since it capitalizes on some forgetting occurring between learning sessions

- Without incentives students are unlikely to use spaced practice on their own — but just as we can learn, they can be coached!



Massed practice (cramming) creates an "illusion of mastery" where information is available in working memory but hasn't been encoded into long-term storage. Spacing introduces a "desirable difficulty"—the slight struggle of recalling something just as it begins to fade—which is what actually makes the learning stick.

Reminder: Spacing establishes the necessary time intervals that make subsequent retrieval attempts more cognitively demanding and therefore more effective for long-term retention. By alternating between different topics or skills, interleaving naturally creates these spaces while forcing the mind to differentiate between various concepts. Together, these strategies ensure that knowledge is deeply integrated and readily accessible across diverse contexts.

For Teachers: Designing the "Spaced" Unit

- The 10-20% Rule: Reserve a small portion of every assessment for old material, ensuring students cannot cram knowledge and then discard it.
- Cumulative "Do-Nows": Use the first five minutes of your 75-minute blocks for retrieval of material from a month ago, not just last night's reading.
- Lagged Homework: Assign practice sets for a topic one week after it was taught. This creates the necessary "forgetting gap" that strengthens long-term circuits.
- Spiral Planning: Map your semester so that core concepts from Unit 1 are explicitly required to solve complex problems in Unit 4.

For Students: Studying for the Long Haul

- Ditch the Marathon: Replace one four-hour "cram" session with four one-hour sessions spread across the week; you'll learn more in the same total time.
- The "Sleep Reset": Study a concept in the evening and quiz yourself the next morning. Sleep helps consolidate the memory you just spaced.
- Use the Leitner System: Mark your calendar to revisit difficult topics at increasing intervals (e.g., 1 day later, then 3 days, then 1 week).
- Avoid the Comfort Trap: If studying feels fast and easy, you're likely just recognizing the info, not learning it; spacing makes it feel harder, which means it's working.